

Chapter 4, Section 5

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1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\text{Aut } X$ of automorphisms of X is finite.

Proof. X is hyperelliptic. Let $\pi : X \rightarrow \mathbb{P}^1$ be the unique g_2^1 of X . By the Riemann-Hurwitz formula, π is ramified at $2g + 2$ points. If σ is an automorphism of X , then there exists $\tau \in \text{PGL}(1)$ such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^1 & \xrightarrow{\tau} & \mathbb{P}^1 \end{array}.$$

This means $\text{Aut}(X)$ permutes the branch points of π , so there is a group homomorphism $\text{Aut}(X) \rightarrow S_{2g+2}$. The latter is a finite group, so it remains to show it has finite kernel. If σ fixes every branch point, then τ is the identity, so σ is either the identity map, or the unique hyperelliptic involution (think in terms of function fields and the fact that the hyperelliptic involution is induced by the unique non-trivial element in $\text{Gal}(K(X)/K(\mathbb{P}^1))$).

X is nonhyperelliptic. Consider the canonical embedding $X \hookrightarrow |K| \cong \mathbb{P}^{g-1}$. If σ is an automorphism of X , then there exists $\tau \in \text{PGL}(g-1)$ such that we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^{g-1} & \xrightarrow{\tau} & \mathbb{P}^{g-1} \end{array}.$$

By (Ex. 4.6), X has $g(g^2 - 1)$ hyperosculation points in this embedding, and τ must permute them because linear transformations preserve incidence structures. So there is a group homomorphism $\text{Aut}(X) \rightarrow S_{g(g^2-1)}$, and since $S_{g(g^2-1)}$ is finite, it remains to show the kernel is finite. For every $g \geq 3$, $X \cap H$ can contain at most point two hyperosculation points for any hyperplane $H \subset \mathbb{P}^{g-1}$, i.e., no three hyperosculation points are collinear. It is a general fact that any automorphism in $\text{PGL}(n)$ is the identity map if it fixes at least $n + 2$ points in general position. Since $g(g^2 - 1) \geq g + 1$ for every $g \geq 3$, if τ fixes the hyperosculation points, it must be the identity. Hence, σ is the identity. As a corollary, any automorphism of a nonhyperelliptic is induced by an automorphism of the canonical embedding, i.e., there exists an injective map $\text{Aut}(X) \rightarrow \text{PGL}(g-1)$. □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g_3^1 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbb{P}^1$, ramified at $0, 1, \infty$, and $2g - 1 = 7$ additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of $\mathfrak{M}_4$ (5.5.5).

The canonical model of a nonhyperelliptic curve of genus 4 is the complete intersection of a unique quadric surface and a cubic surface in $\mathbb{P}^3$ (5.2.2). The family of all these curves is parametrized by the space $\mathfrak{S}$

of all complete intersections of a quadric surface and a cubic surface. We can think of $\mathfrak{S}$ as the set of all homogenous ideals generated by a degree two and three homogenous polynomial in $k[x_1, \dots, x_4]$. So there is a morphism $\mathfrak{S} \rightarrow \mathfrak{M}_4$, whose fibres are images of the group $\mathrm{PGL}(3)$, which has dimension 15. It remains to compute the dimension of $\mathfrak{S}$. There is a projection map $\mathfrak{S} \rightarrow \mathfrak{Q}$, where $\mathfrak{Q}$ is the space of all irreducible quadric surfaces in $\mathbb{P}^3$. It is a locally closed subset of $|\mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))|$, which has dimension $\binom{3+2}{2} - 1 = 9$. The fibres of $\mathfrak{S} \rightarrow \mathfrak{Q}$ is the image of pullback of the linear system of cubic surfaces in $\mathbb{P}^3$, which is just $|\mathrm{H}^0(Q, \mathcal{O}_Q(3))|$ for some quadric surface $Q \subset \mathbb{P}^3$. Note that the dimension of this space is independent of the choice of Q . Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_Q \longrightarrow 0.$$

Tensoring by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) \longrightarrow \mathrm{H}^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow \mathrm{H}^0(Q, \mathcal{O}_Q(3)) \longrightarrow 0.$$

By exactness, we have

$$\dim |\mathrm{H}^0(Q, \mathcal{O}_Q(3))| = \binom{3+3}{3} - \binom{3+1}{1} - 1 = 15,$$

which implies the fibres of $\mathfrak{S} \rightarrow \mathfrak{Q}$ have dimension 15, so $\mathfrak{S}$ has dimension $9 + 15 = 24$. Hence, $\mathfrak{M}_4$ has dimension $24 - 15 = 9$. The subspace of $\mathfrak{Q}$ consisting of quadric cones have dimension 8, so the space of these curves have codimension one in $\mathfrak{M}_4$. Hence, the subset of nonhyperelliptic curves having only one g_3^1 is an irreducible family of dimension 8. $\square$

5. Curves of Genus 5. Assume X is not hyperelliptic.

- (a) The curves of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection $F_2.F_2.F_2$ form a family of dimension 12.
- (b) X has a g_3^1 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.
- (c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbb{P}^2 \rightarrow \mathbb{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbb{P}^4$, with V isomorphic to $\mathbb{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g_3^1 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X . Thus V and g_3^1 are unique.

Proof.

- (a) A curve of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection of three quadrics is given by a three-dimensional subspace of $\mathrm{H}^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2))$, which has dimension $\binom{4+2}{2} = 15$. Thus, the family of all these curves is parametrized by an open subset of the Grassmannian $\mathrm{Gr}(3, 15)$, which has dimension $3(15 - 3) = 36$. There is a natural action of $\mathrm{PGL}(4)$ on $\mathrm{Gr}(3, 15)$, and since the embedding is canonical, two three-dimensional subspaces in $\mathrm{Gr}(3, 15)$ give isomorphic curves if and only if they are in the same orbit of $\mathrm{PGL}(4)$. So there is a morphism $\mathrm{Gr}(3, 15) \dashrightarrow \mathfrak{M}_5$, whose fibres are the images of the group $\mathrm{PGL}(4)$ which has dimension $(4 + 1)^2 - 1 = 24$. Since any individual curve has only finitely many automorphisms (Ex. 5.2), the fibres have dimension $= 24$, and so the image of $\mathrm{Gr}(3, 15)$ has dimension $36 - 24 = 12$.
- (b) A canonical curve X of genus 5 in $\mathbb{P}^4$ has a g_3^1 if and only if it has a trisecant. This means there exists distinct points there exists a divisor D of degree 3 such that $\dim |K - D| = 2$ (5.5.3). The complete linear system $|K - D|$ is base-point free, since otherwise if P is a base-point, then

$$\dim |D + P| = \dim |K - D - P| = 2,$$

which means $|D + P|$ is a g_4^2 on X . This is impossible for a nonhyperelliptic curve of genus 5, since the image is either a conic or a quartic curve. The former corresponds to a g_2^1 on X , and the latter implies X has genus at most $\binom{4-1}{2} = 3$. Therefore, $X \rightarrow |K - D| \cong \mathbb{P}^2$ must be birational by reasons of degree, and

by (3.11.1) the image is a plane quintic with a singular point of multiplicity 2, which is either a cusp or a node.

Conversely, suppose X can be represented as a plane quintic $X' \subseteq \mathbb{P}^2$ with one node at $P \in X'$. Let $\pi : V \rightarrow \mathbb{P}^2$ be the blowing up of $\mathbb{P}^2$ with center at P , and let $E \subset V$ be the exceptional divisor of π . Let H be any hyperplane in $\mathbb{P}^2$ so that $|2H|$ is the linear system of conics in $\mathbb{P}^2$. Using the results of (V, §3) and (V, §4), the linear system $|2H - P|$ of conics passing through P has dimension $5 - 1 = 4$ with an assigned basepoint at P . By (II, 7.17.3), the rational map $\mathbb{P}^2 \dashrightarrow |2H - P| \cong \mathbb{P}^4$ extends to a morphism $V \rightarrow \mathbb{P}^4$. It is an embedding of V into $\mathbb{P}^4$ because it corresponds to the linear system $|D|$ with $D = 2\pi^*H - E$ on V , which is very ample by (V, 4.2.2). The strict transform $\tilde{X}'$ of X' is nonsingular, i.e., $\tilde{X}' \cong X$, and by (V, 3.6), it relates to X' as divisors in V in the following way

$$\pi^*X' = \tilde{X}' + 2E.$$

Using the fact that $X' \sim 5H$ in $\mathbb{P}^2$, by (V, Ex. 1) the degree of $\tilde{X}'$ as a curve in $\mathbb{P}^4$ equals to

$$\tilde{X}' \cdot D = (5\pi^*H - 2E) \cdot (2\pi^*H - E) = 10(\pi^*H)^2 + 2E^2 = 10 - 2 = 8.$$

By the Riemann-Roch theorem, the only degree $2g - 2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf, so this embedding of X in $\mathbb{P}^4$ is canonical. Now, the identification of V with a rational scroll $V \rightarrow \mathbb{P}^1$ of degree 3 in $\mathbb{P}^4$ (V, 2.19.1) induces the following correspondence of divisors on V

$$a\pi^*H - bE = (a - b)C_0 + af,$$

where C_0 is a section and f is any fibre of the ruling $V \rightarrow \mathbb{P}^1$. Rewriting $\tilde{X}'$ in this way gives

$$\tilde{X}' = 3C_0 + 5f.$$

Since $\tilde{X}' \cdot f = 3$, a fibre of $V \rightarrow \mathbb{P}^1$ and $\tilde{X}'$ intersect at three points. Also, V is a rational scroll in $\mathbb{P}^4$, so any fibre of V is a line in $\mathbb{P}^4$. Hence, any fibre of $V \rightarrow \mathbb{P}^1$ is a trisecant of X , so X has a 1-parameter family of trisecants in $\mathbb{P}^4$.

By (3.11.1), the space $V_{d,r}$ of irreducible plane curves of degree d and r nodes is irreducible and has dimension $\frac{1}{2}d(d+3) - r$. In our case, $V_{5,1}$ has dimension 19, and we have just shown $V_{5,1}$ parametrizes the space of all genus 5 curves with a g_3^1 . So there is a morphism $V_{5,1} \rightarrow \mathfrak{M}_5$, whose fibres are images of the group $\mathrm{PGL}(2)$, which has dimension 8. Hence, the family of genus 5 curves with a g_3^1 form an irreducible family of dimension $19 - 8 = 11$.

(c) The results of this exercise combined with (5.5.3) can be summarized by the commutative diagram

$$\begin{array}{ccccc} & & \mathbb{P}^2 & & \\ & \nearrow \text{plane quintic} & \uparrow \text{blow up} & & \\ X & \hookrightarrow & V & \hookrightarrow & \mathbb{P}^4 \cong |K| \cong |D| \\ & \searrow g_3^1 & \downarrow \text{ruled surface} & & \\ & & \mathbb{P}^1 & & \end{array}$$

where K is a canonical divisor of X , and D is the divisor of V from (Ex. 5.5b). Thus, we have three different representations of nonhyperelliptic curves of genus 5 with a g_3^1 .

□

6. Show that a nonsingular plane curve of degree 5 has no g_3^1 . Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Proof. Let $X \subset \mathbb{P}^2$ be a nonsingular plane curve of degree 5. The Veronese embedding $\mathbb{P}^2 \xrightarrow{\sim} V \subset \mathbb{P}^5$ realizes $\mathbb{P}^2$ as a surface of degree 4, and realizes X as a curve of degree 10 in $\mathbb{P}^5$. Since $g = 6$, this embedding is the canonical embedding. In particular, the only degree $2g - 2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf. The proof of (5.5.3) applies to any nonhyperelliptic curve. In particular, X has a g_3^1 if and only if its canonical embedding in $\mathbb{P}^5$ has a trisecant line. Suppose that $L \subset \mathbb{P}^5$ is a trisecant line to the canonical image of X , meeting X at three points P, Q, R . Since $X \subset V$, the line L meets the Veronese surface in P, Q, R . These three points determine the g_3^1 on X , and

hence $\dim |P + Q + R| = 1$. On the other hand, three points of the Veronese surface that are collinear in $\mathbb{P}^5$ must be collinear in $\mathbb{P}^2$, because the line through P, Q, R intersects V at three points, and V has degree 2 in $\mathbb{P}^5$, so the line must be contained in V . Thus P, Q, R are collinear in $\mathbb{P}^2$. But there is only one line in $\mathbb{P}^2$ passing through three distinct collinear points, contradicting the dimension of $|P + Q + R|$. Therefore, X has no trisecant line in its canonical embedding, and consequently X has no g_3^1 .

For the second part, it is enough to show there exists nonhyperelliptic curves of genus 6 with a g_3^1 . Consider any irreducible nonsingular curve X associated to a divisor of type $(3, 4)$ on the nonsingular quadric surface Q . It has genus $(3 - 1)(4 - 1) = 6$, and one of the families of lines on Q cuts out a g_3^1 on X . Hence, X cannot be represented as a nonsingular plane quintic curve. The canonical sheaf of Q is $\omega_Q = \mathcal{O}_Q(-2, -2)$, so by (II, 8.20), the canonical sheaf of X is $\omega_X \cong \mathcal{O}_Q(1, 2) \otimes \mathcal{O}_X$. Since $\mathcal{O}_Q(1, 2)$ is a very ample invertible sheaf on Q , ω_X is very ample as well. Hence, X is not hyperelliptic by (5.2). $\square$